FYUGP B.Sc. PHYSICS HONOURS

ELECTRODYNAMICS II

Semester V · Core in Major

MODULE 4: TRANSIENT CIRCUITS AND ALTERNATING CURRENTS

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

Transient currents in L-R circuits

1 Write the instantaneous current in an L-R circuit after the cell is removed and the circuit is closed on itself, if the current at the instant of removal is I_0. **(1 mark)**

Oscillations of an L-C circuit

2 Write the natural frequency ν of an L-C circuit in terms of L and C. **(1 mark)**

EMF in a rotating coil

3 Write the peak value E_0 of the emf induced in a coil of N turns and area A rotating with angular velocity ω in a uniform field B. **(1 mark)**

Phase relations

4 State whether true or false: in a purely inductive circuit the current leads the applied emf by $90°$. **(1 mark)**

5 Write the impedance Z of a series R–C circuit in terms of R, C and ω. **(1 mark)**

The j operator

6 Write the impedance of an inductor L in complex (j) notation. **(1 mark)**

Section B: 2 marks each, answer all

Mean value of alternating current

7 Write the mean value of an alternating current over half a cycle in terms of its peak value I_0, and the r.m.s. value in terms of I_0. **(2 marks)**

Phase angle of a series R–C circuit

8 State whether the current leads or lags the applied emf in a series R–C circuit. Write the expression for the phase angle. **(2 marks)**

Parallel resonance

9 Write the resonant frequency of a parallel LC circuit in which the resistance of the coil is negligible. **(2 marks)**

Series resonance

10 Sketch the current against frequency for a series LCR circuit with a low and a high value of R. Mark the resonant frequency on the graph. **(2 marks)**

Power in a resistive AC circuit

11 Write the average power consumed in a purely resistive AC circuit in terms of the r.m.s. emf and r.m.s. current. **(2 marks)**

Section C: 4 marks each, answer any two

Growth of current in L-R

12 A coil of inductance 0.5 H and resistance 10 Ω is connected to a 12 V cell. Find the time constant, the steady current, and the current after 0.1 s. Take $e^{-2} = 0.1353$. **(4 marks)**

Charging and discharge of a capacitor

13 A capacitor of 10 μF is charged to 100 V and then discharged through a resistance of 1 MΩ. Find the initial charge, the charge remaining after 10 s, and the time in which the charge falls to $1/e$ of its initial value. **(4 marks)**

Parallel resonant circuit

14 A coil of inductance 0.1 H is connected in parallel with a capacitor of 4 μF. Find the resonant frequency of the circuit, taking the resistance of the coil to be negligible. **(4 marks)**

Power factor and true power

15 A coil of resistance 30 Ω and inductance 0.1 H is connected to a 240 V, 50 Hz supply. Find the impedance, the power factor, and the true power. **(4 marks)**

Section D: 10 marks each, answer any one

Series LCR circuit: phasor (j) method

16 A series LCR circuit is driven by $E = E_0 e^{j\omega t}$. Using the j operator, obtain the impedance Z, the current I, and the phase angle ϕ. Show that the current amplitude is a maximum at $\omega = 1/\sqrt{LC}$, and state the peak value of the current there. **(10 marks)**

Parallel LC circuit

17 A coil of inductance L and resistance R is connected in parallel with a capacitance C across an AC source. Show that the impedance of the parallel combination is a maximum at $\omega_0 = 1/\sqrt{LC}$ when R is negligible. Explain why this is called a rejector circuit. **(10 marks)**

MODULE 4: MODEL ANSWERS

For checking your work. Each answer shows the key step and the result.

Section A. A1 $I = I_0 e^{-Rt/L}$ (Murugeshan §12.2, Eq. 2). A2 $\nu = 1/(2\pi\sqrt{LC})$ (§12.5, Eq. 4). A3 $E_0 = NBA\omega$ (§13.1). A4 False: in a pure inductor the current lags the emf by $\pi/2$ (§13.2(ii)). A5 $Z = \sqrt{R^2 + 1/(\omega C)^2}$, from $Z = R + 1/(j\omega C)$ (§13.2(v)). A6 $Z_L = j\omega L$ (§13.2 j operator).

Section B. B7 Mean over half a cycle $I_{mean} = 2I_0/\pi = 0.637\, I_0$; r.m.s. $I_{rms} = I_0/\sqrt{2} = 0.707\, I_0$ (§13.1). B8 The current leads the emf by $\theta = \tan^{-1}\dfrac{1/(\omega C)}{R}$ (§13.2(v), Eq. 5). B9 $\nu_0 = 1/(2\pi\sqrt{LC})$ (§13.4). B10 Both curves peak at ν_0 with height E_0/R; the low-R curve is taller and sharper (§13.3, Figs. 13.17–13.18). B11 $P = E_{rms} I_{rms}$, since $\cos\phi = 1$ (§13.5, special case 1).

Section C. C12 Time constant $L/R = 0.05$ s; steady current $I_0 = 12/10 = 1.2$ A; $I(0.1) = 1.2(1 - e^{-2}) = 1.2 \times 0.8647 = 1.04$ A (§12.1, Eq. 4). C13 $Q_0 = CV = 10^{-5} \times 100 = 1.0 \times 10^{-3}$ C; $CR = 10$ s; $Q(10\text{ s}) = Q_0 e^{-1} = 3.68 \times 10^{-4}$ C; the charge falls to $1/e$ at $t = CR = 10$ s (§12.3(b), Eq. 2). C14 $\nu_0 = \dfrac{1}{2\pi\sqrt{(0.1)(4 \times 10^{-6})}} = 251.6$ Hz, about 252 Hz (§13.4, Eq. 2). C15 $X_L = 2\pi(50)(0.1) = 31.42\ \Omega$; $Z = \sqrt{30^2 + 31.42^2} = 43.44\ \Omega$; $\cos\phi = 30/43.44 = 0.691$; $I = 240/43.44 = 5.52$ A; $P = E\,I\cos\phi = 916$ W (§13.5).

Section D. D16 $Z = R + j(\omega L - 1/\omega C)$, so $|Z| = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$ and $\tan\phi = (\omega L - 1/\omega C)/R$. The current is $I = E/Z = I_0 e^{j(\omega t - \phi)}$ with $I_0 = E_0/|Z|$. The amplitude is largest when $\omega L = 1/\omega C$, that is $\omega = 1/\sqrt{LC}$, where $|Z| = R$ and $I_0 = E_0/R$ (§13.3, second method, Eqs. 2–5). D17 For the parallel branch $1/Z = 1/(R + j\omega L) + j\omega C$. With $R \to 0$, $1/Z = j(\omega C - 1/\omega L)$, which is zero at $\omega_0 = 1/\sqrt{LC}$, so Z is infinite there. The circuit rejects the current of its natural frequency, which is why it is called a rejector (§13.4).

Mapping to the syllabus and textbook

Syllabus item	Topic	Murugeshan section	Model questions
15	Growth of current in L-R, C-R and L-C circuits	12.1, 12.3, 12.5	1, 12, 21, 22
16	Decay of current in L-R, C-R and L-C circuits	12.2, 12.3, 12.5	1, 2, 13
17	EMF induced in a rotating coil; mean and r.m.s. values	13.1	3, 7, 16
18	AC circuits with R, L and C alone; L-R and C-R	13.2	4, 5, 8, 14
19	Use of the j operator	13.2	6, 16
20	Parallel L and C; parallel resonance	13.2, 13.4	5, 9, 14, 17
21	Series LCR circuit and resonance	13.3	10, 15, 16
22	Power in AC circuits; power factor	13.5	11, 15